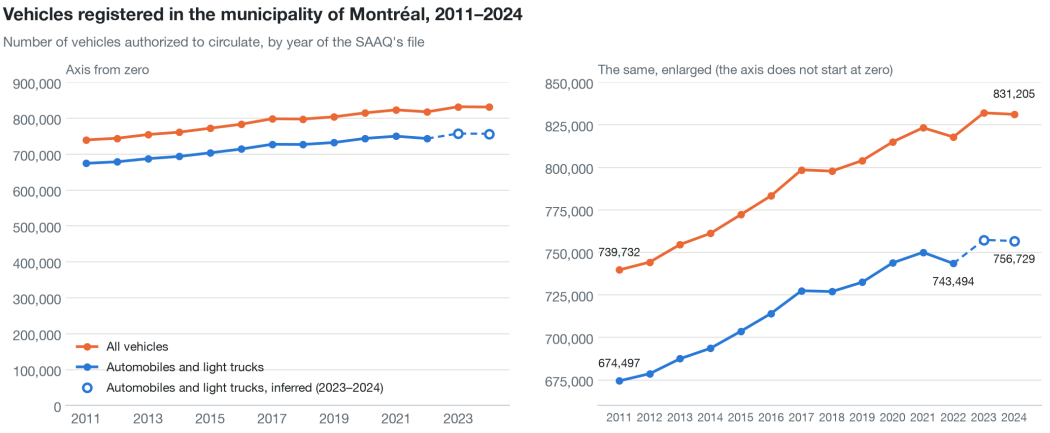
Vehicle registrations in Québec, 2011–2024

Joining the SAAQ's open data with the files released under the access Act

Rick Hoge, PhD

Draft, November 10, 2026 v2026-11-10-23-11-41

Overview: the municipality of Montréal, 2011–2024



Type is not in the 2023–2024 files: it is inferred from make, model and model year. Applied to each published year 2012–2022 from the years before it, the method gives +0.05 % to +0.13 % against the published count (21 tests).
Vehicles left unassigned: 2,432 (2023), 2,917 (2024).
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

Two sources, one row per registered vehicle

Years	Source	Columns
2011–2022	SAAQ open data, « Véhicules en circulation » (CC BY 4.0), files as revised by the SAAQ on December 18, 2023	class, type, make, model, model year, net mass, cylinders, displacement, axles, colour, fuel (from 2017), region, MRC, municipality code
2023–2024	Files released by the SAAQ under the access Act, file 03.06.34729 (decision of September 17, 2025)	year, make, model, model year, municipality name, region, first three characters of the postal code (FSA)

- Rows: 5,985,463 in 2011, 7,058,605 in 2022, 7,169,989 in 2023, 7,189,594 in 2024.
- The portal holds no year after 2022. Its note: « Les données ont été révisées pour respecter les nouvelles dispositions de la loi 25 ».
- The decision withheld the other columns under section 53 of the Act.

What the two sources share

- **Shared:** make, model, model year, and the municipality.
- **Only in 2011–2022:** class, type, fuel, net mass. A type or fuel figure for 2023–2024 is therefore **inferred**, and is labelled so wherever it appears.
- **Only in 2023–2024:** the FSA. A figure by FSA exists for two years only.
- **Not inferred for 2023–2024:** the class (personal or commercial use). It describes the use made of a vehicle, not the vehicle, so make, model and model year do not settle it.

Step 1: one vocabulary of makes and models

- The open data write makes and models as codes of at most five characters (TOYOT, COROL). The release mixes those with full names, other abbreviations and misspellings: more than 7,000 make strings against 418 codes.
- As released, 80.9 % of the 2023 vehicles and 75.6 % of the 2024 vehicles carry a make and model in the open data's own main spelling.
- Stated rules (first five characters, punctuation, two short alias tables, unambiguous prefixes, trim suffixes) bring this to 97.1 % and 96.8 %. A further 2.4 % and 2.7 % resolve to the make only; 0.4 % and 0.5 % stay unresolved. Nothing is matched by similarity.
- Some codes mean other vehicles in the release than in the open data (DODGE CARAV, merged into GRAND by the curators). They are detected from vehicles of model year 2021 or earlier, whose number per code cannot grow after 2022; 52 codes (148,434 vehicles in 2023) are reassigned by model-year profile.
- Municipality names are linked to the codes of 2011–2022 for 98.3 % of the 2024 vehicles (ISQ geographic codes, July 31, 2025).

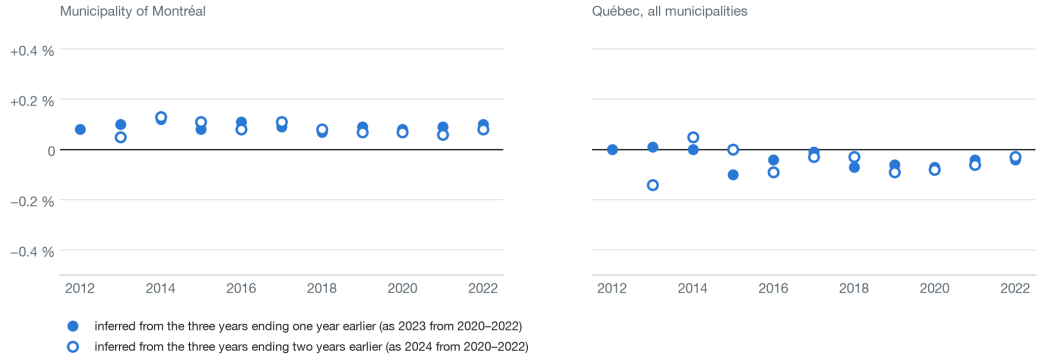
Step 2: type and fuel as expected counts

- The open data of 2020–2022 give, for each make, model and model year, how many vehicles are of each type and each fuel.
- A vehicle of 2023–2024 is not given one value. It contributes to each value in the proportions observed for its key.
- The key is the most specific one found: make, model and model year (94.8 % of the 2023 vehicles, 88.3 % of 2024); then make and model on recent model years (2.0 %, 8.2 %); then make and model, make and model year, make.
- Vehicles matching no key stay unassigned: 0.5 % in 2023, 0.5 % in 2024.
- **The method is tested where the answer is known:** each published year is reduced to the release's seven columns, inferred from earlier years, and compared with what the SAAQ published.

The test on published years: vehicle type

Inferring vehicle type from make, model and model year: error on published years

Automobiles and light trucks: inferred count against the count the SAAQ published, in per cent



Each published year is reduced to the seven columns of the 2023–2024 files, inferred, and compared with the published type. The test uses the open data's own make and model codes; it does not measure the effect of the unedited codes of 2023–2024.
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

The test on published years: what it shows

Quantity inferred	Difference from the published count
Automobiles and light trucks, Montréal (21 tests)	$+0.05$ % to $+0.13$ %
Automobiles and light trucks, Québec	-0.14 % to $+0.05$ %
Same, by municipality, 2022: under 1,000 vehicles	median 0.58 %
Same, by municipality, 2022: 10,000 to 99,999	median 0.11 %
Gasoline, Québec, one year ahead	-0.11 % to 0.11 %
Electric, Québec, one year ahead	-7.7 % to -3.8 %
Electric, Québec, two years ahead	-18.8 % to -12.4 %
Plug-in hybrid, Québec, one year ahead	-25.0 % to $+11.1$ %

- Type is recovered to a fraction of one per cent.
- Electric vehicles are under-counted, because the newest model years are missing from the reference. No electric count for 2023–2024 is given here.
- The test uses the open data's own codes. It does not measure the effect of the release's unedited codes.

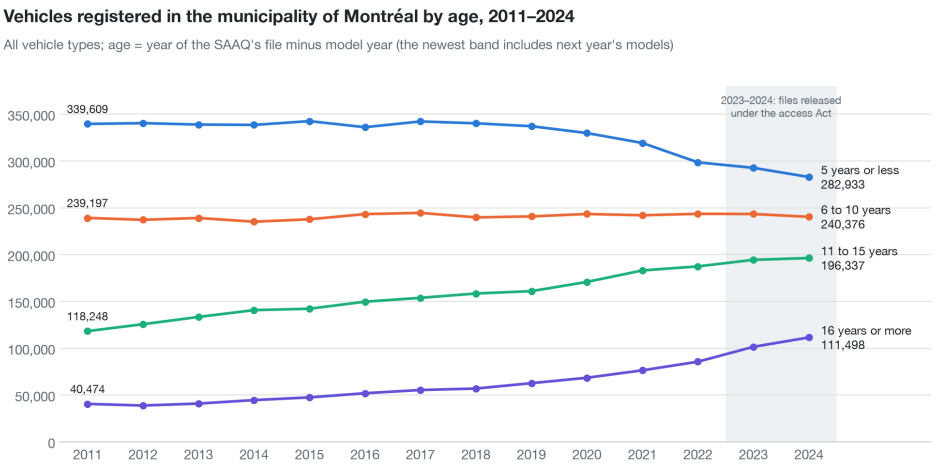
Result: Montréal, automobiles and light trucks

Municipality of Montréal	2011	2021	2022	2023	2024
All vehicles (released)	739,732	823,314	817,880	832,054	831,205
Automobiles and light trucks	674,497	750,071	743,494	*757,376*	*756,729*

Italic: inferred. Released: SAAQ open data (files of December 18, 2023) to 2022; access release 03.06.34729 for 2023–2024.

- 2011 to 2021: 75,574 more automobiles and light trucks (+11.2 %).
- 2022 is lower than 2021 by 6,577 in the published files; all vehicles fall by 5,434 the same year.
- 2023 and 2024 (inferred) stand 7,305 and 6,658 above 2021; the two years differ by 647.

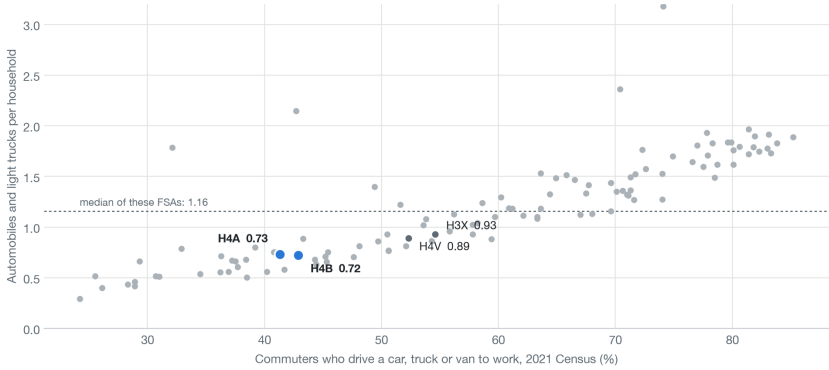
Vehicles by age: nothing inferred



Automobiles per household by FSA, 2024

Automobiles and light trucks per household, by FSA: island of Montréal and Laval, 2024

Each dot is one FSA beginning with H (113 with 1,000 households or more); rue de Terrebonne runs through H4A and H4B



Vehicles: 2024 file, type inferred from make, model and model year; all owners, businesses included, so an FSA with fleets or dealers stands high. Households and commuting: 2021 Census, dissemination areas shared among FSAs by postal codes (a few per cent of uncertainty). The two quantities are three years apart. One FSA is above the scale (H2Z, 4.67). Statistics Canada, 2021 Census, catalogue 98-401-X2021006; postal codes: Service Objects (March 2024) and DMTI Spatial, CanMap Postal Code Suite v2024.3.
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

The FSAs around rue de Terrebonne

FSA	Automobiles and light trucks, 2024 (inferred)	Households, 2021 Census	Per household	Commuters who drive (%)	Rank from the lowest, FSAs in H
H4A	9,598	13,141	0.73	41.3	27 of 113
H4B	7,351	10,193	0.72	42.9	26 of 113
H4V	8,199	9,205	0.89	52.3	41 of 113
H3X	9,999	10,747	0.93	54.6	43 of 113

- Rue de Terrebonne runs through H4A and H4B. The median of the 113 FSAs is 1.16 per household.
- The vehicle count includes vehicles owned by businesses. The ratio is an average: it does not say how many households have no vehicle.
- Households come from the 2021 Census by dissemination area, shared among FSAs by postal codes (a few per cent of uncertainty); vehicles are of 2024.

Limits, and what is not yet done

- Type and fuel for 2023–2024 are inferred; class cannot be. Electric and plug-in counts for 2023–2024 are not given.
- By FSA there are two years, so no local trend.
- Registrations count vehicles owned, not trips made or distance driven.
- Not yet done: the remaining charts of the 2025 application checked against this pipeline; fleet mass and the Road Wear Index (2011–2022); scenarios for vehicle storage; driver's licences; the written note; the French versions.
- Code, tables and figures:
  `github.com/OpenMobilityData/Quebec-vehicle-registrations`